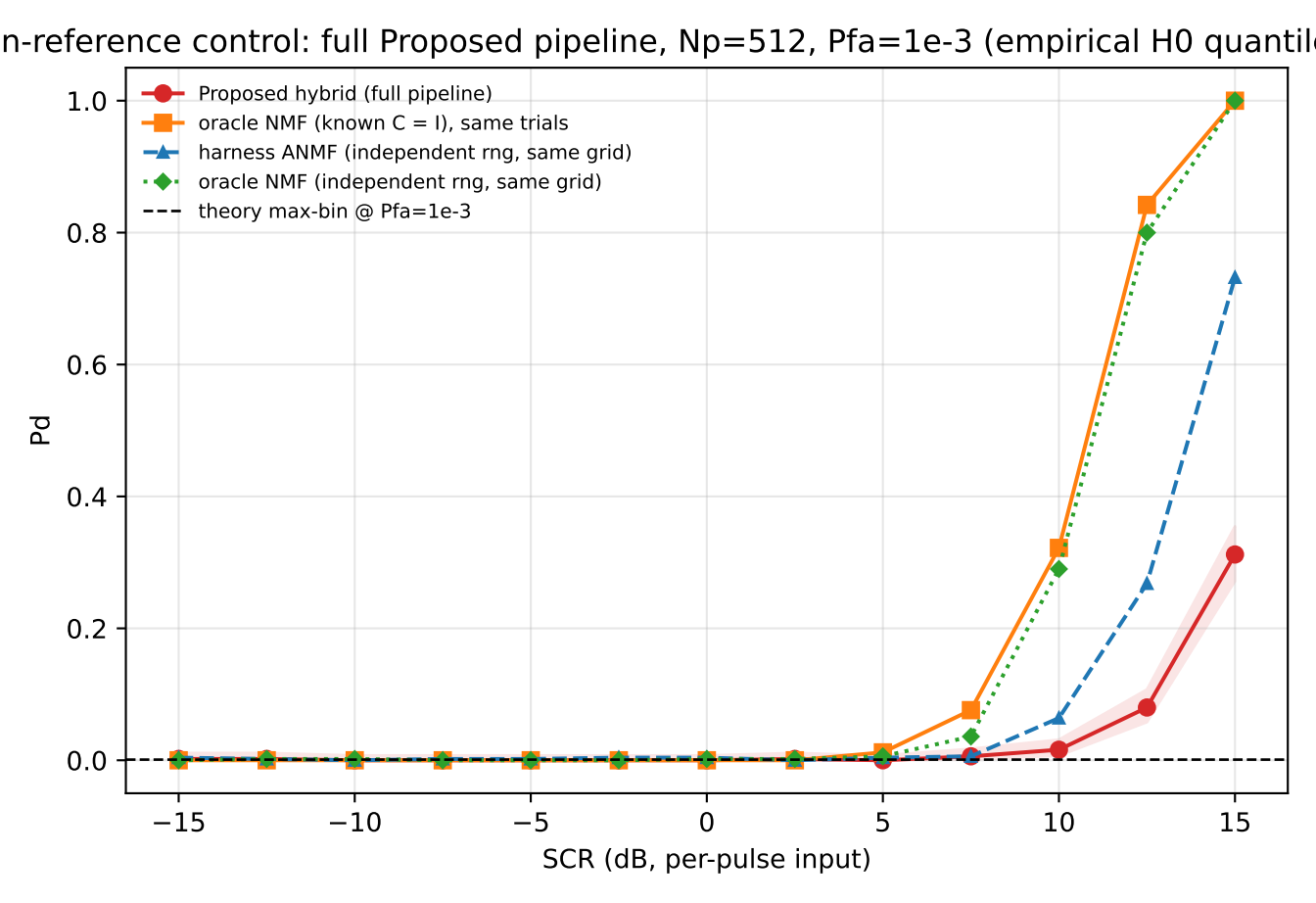
n-reference control: full Proposed pipeline, Np=512, Pfa=1e-3 (empirical H0 quantil
Proposed hybrid (full pipeline)
oracle NMF (known C = I), same trials
harness ANMF (independent rng, same grid)
oracle NMF (independent rng, same grid)
theory max-bin @ Pfa=1e-3
Pd
1.0
0.8
0.6
0.4
0.2
0.0
−15
−10
−5
0
5
10
15
SCR (dB, per-pulse input)